

Def. Let  $p: E \rightarrow B$  be a continuous onto map.

An open set  $U \subset B$  is said to be evenly covered by  $p$  if:

$$p^{-1}[U] = \bigcup_{\alpha \in I} V_{\alpha} \text{ where } V_{\alpha} \text{ is disjoint open sets such that}$$
$$\text{for all } \alpha \in I, \quad U \stackrel{p}{\cong} V_{\alpha}$$

The map  $p: E \rightarrow B$  is called a covering map if:

for each  $b \in B$ , there exists an evenly covered neighbor  $U$  containing  $b$ .

If such a map  $p$  exists, then  $E$  is called the covering space.

Prop. Every covering map is open map.

Proof. Let  $p: E \rightarrow B$  be a covering map, and  $W \subset E$  be an open set.

To show  $p[W]$  is an open set, let  $b \in p[W]$  be given.

Since the  $p$  is covering map, there exists an evenly covered open set  $U$  containing  $b$ . By definition, for some disjoint open sets  $V_{\alpha}$  ( $\alpha \in I$ ),

$$p^{-1}[U] = \bigcup_{\alpha \in I} V_{\alpha} \quad b \in U$$

and,  $b \in p[W]$ , so  $b = p(y)$  for some  $y \in W$ . Since  $b \in U$ , so  $p^{-1}[p[\{y\}]] \subset p^{-1}[U]$ , therefore  $y \in p^{-1}[U] = \bigcup_{\alpha \in I} V_{\alpha}$ , thus  $y \in V_{\alpha_0}$  for some  $\alpha_0 \in I$ .

Now,  $V_{\alpha_0} \cap W$  is an open set of  $E$  containing  $y$ , so that  $V_{\alpha_0} \cap W$  is an open set of  $V_{\alpha_0}$ .

Since the restriction

$$p|_{V_{\alpha_0}}: V_{\alpha_0} \rightarrow U$$

is Homeomorphism, so  $p[V_{\alpha_0} \cap W] \subset U$  is an open set of  $U$ , and since  $U$  is open set of  $B$ , thus  $p[V_{\alpha_0} \cap W]$  is open set of  $B$ .

Finally,  $b = p(y) \in p[V_{\alpha_0} \cap W] \subset p[W]$ , so  $p[W]$  is an open set of  $B$ .  $\square$

Lift.

Def. Let  $p: E \rightarrow B$  be a Continuous map.

For a Continuous map  $f: X \rightarrow B$ , a lifting of  $f$  is:

a Continuous map  $\tilde{f}: X \rightarrow E$  such that  $p \circ \tilde{f} = f$ .



Lemma. Let  $p: E \rightarrow B$  be a Covering map and  $f: X \rightarrow B$  be Continuous maps.

If there exist two lifts  $\tilde{f}_1, \tilde{f}_2: X \rightarrow E$  of  $f$  with  $\tilde{f}_1(x_0) = \tilde{f}_2(x_0)$  for some  $x_0 \in X$ , and  $X$  is Connected, then  $\tilde{f}_1 = \tilde{f}_2$ .

Thm. Let  $p: (E, e_0) \rightarrow (B, b_0)$  be a covering map.

For any Path  $\alpha: I \rightarrow B$  such that  $\alpha(0) = b_0$ , there exists a Unique Path

$\tilde{\alpha}: I \rightarrow E$  such that  $p \circ \tilde{\alpha} = \alpha$  and  $\tilde{\alpha}(0) = e_0$ .

Diagrammically,

